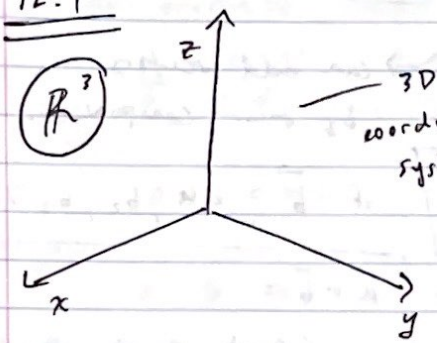
MATH 251 — TEST 1 REVIEW

2/4/2020

12.1

$\mathbb{R}^3$



— 3D coordinate system

* projecting points onto a plane —

e.g. $(1,2,3)$ onto xz plane:

↳ $y = 0$

↳ $(1,0,3)$

$\left.\begin{matrix} x \\ y \\ z \end{matrix}\right\} = k$, some constant

↳ represent PLANES

distance formula:

$$|P_1P_2| = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2}$$

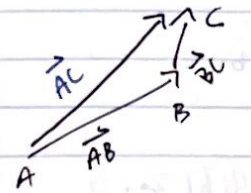
12.2

$\vec{v}$ or $\mathbf{v}$

vector

↳ magnitude & direction
(length) (angle)

combining vectors —



$\vec{AC} = \vec{AB} + \vec{BC}$

tail ↳ head

sphere math. definition:

all points $P(x,y,z)$ whose distance from C is r.

↳ $C(h,k,l)$

$$(x-h)^2 + (y-k)^2 + (z-l)^2 = r^2$$

vector from two points —

$A\langle x_1, y_1, z_1\rangle$ and $B\langle x_2, y_2, z_2\rangle$

$\vec{AB}$ = head − tail =

$\langle x_2-x_1, y_2-y_1, z_2-z_1\rangle$

scalar multiples —

given SCALAR c and VECTOR v, scalar multiple

$\boxed{cv = c \cdot |v|,}$ same directions as v

↳ UNLESS $c < 0$!!

length / magnitude of a vector —

given $\vec{a} = \langle a_1, a_2, a_3 \rangle$ → can add vectors by their components →

$$\boxed{\|a\| = \sqrt{a_1^2 + a_2^2 + a_3^2}}$$

if $\vec{b} = \langle b_1, b_2, b_3 \rangle$

$$\boxed{\vec{a} + \vec{b} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle}$$

Standard basis vectors —

$\underline{\underline{i, j, k}}$ → UNIT VECTORS: a vector that's length = 1 (i, j, k)

$\langle 1,0,0 \rangle$ $\langle 0,1,0 \rangle$ $\langle 0,0,1 \rangle$

$$\text{Unit}_a = \frac{\vec{a}}{\|a\|}$$

↳ magnitude of $\vec{a}$

length / magnitude = $\underline{\underline{1}}$

direction $i, j, k = x, y, z$

12.3

"The Dot Product"

given $\vec{a} = \langle a_1, a_2, a_3 \rangle$, $\vec{b} = \langle b_1, b_2, b_3 \rangle$, $a \cdot b =$

$$a_1 b_1 + a_2 b_2 + a_3 b_3 = \|a\| \|b\| \cos\theta$$

angle b/t $\vec{a}$ and $\vec{b}$

"perpendicular / orthogonal"

↳ if $\theta = \frac{\pi}{2}$ → dot product $= \underline{\underline{0}}$

↳ dot product MEANING:

measuring the EXTENT to which $\vec{a}$ and $\vec{b}$ point in the same direction —

$a \cdot b > 0$ WHEN $\theta < \frac{\pi}{2}$ ↳ same direction

$a \cdot b < 0$ WHEN $\theta > \frac{\pi}{2}$ ↳ diff directions

$a \cdot b = 0$ WHEN $\theta = \frac{\pi}{2}$ ↳ perpendicular

Direction angles —

↳ given vector $\vec{a}$

α — angle b/t $\vec{a}$ & x-axis

β — angle b/t $\vec{a}$ & y-axis

γ — angle b/t $\vec{a}$ and z axis

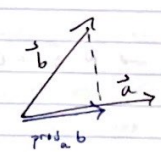
$$\left.\begin{aligned} a &= \langle a_1, a_2, a_3 \rangle \\ &= \langle |a|\cos\alpha, |a|\cos\beta, |a|\cos\gamma \rangle \\ &= |a| \langle \cos\alpha, \cos\beta, \cos\gamma \rangle \end{aligned}\right\}$$

↳ so...

$$\frac{\vec{a}}{|a|} = \underbrace{\langle \cos\alpha, \cos\beta, \cos\gamma \rangle}_{\text{DIRECTIONS}}$$

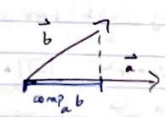
unit vector → length = $\underline{\underline{1}}$

→ PROJECTIONS



vector projection (shadow)

$$\boxed{\text{prod}_a b = \frac{a \cdot b}{|a|^2} a}$$



Scalar projection (magnitude / length of vector projection)

$$\boxed{\text{comp}_a b = \frac{a \cdot b}{|a|}}$$

$$\boxed{\text{WORK} = F \cdot D = \|F\| \|D\| \cos\theta}$$

12.4

"The Cross Product"

given $\vec{a}$: $\langle a_1, a_2, a_3 \rangle$
$\vec{b}$: $\langle b_1, b_2, b_3 \rangle$

↳ direction of $\vec{N}$ determined by right-hand rule (from $\vec{a}$ to $\vec{b}$)

$$\vec{a} \times \vec{b} = \begin{vmatrix} i & j & k \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= (a_2 b_3 - b_2 a_3)\, i - (a_1 b_3 - b_1 a_3)\, j + (a_1 b_2 - b_1 a_2)\, k$$

$\vec{a} \times \vec{b}$ results in a vector $\vec{N}$, "normal" vector that is ORTHOGONAL to BOTH $\vec{a}$ and $\vec{b}$

↳ $\vec{a} \times \vec{b} = |a||b| \sin\theta$

↳ angle b/t a and b

torque : $\tau = r \times F = |r||F| \sin\theta$

↳ vector

↳ positive ↺

SUMMARIZE $\vec{a} \times \vec{b}$ —

↳ vector $\vec{N}$ perpendicular to both $\vec{a}$ and $\vec{b}$ ⟶ $\vec{N} \cdot \vec{a}$ and $\vec{N} \cdot \vec{b} = 0$

↳ orientation determined by right hand rule ($a \to b$)

↳ length $= |a||b| \sin\theta$ ⟶ $\vec{a} \times \vec{b}$ = area of parallelogram with $\vec{a}$ and $\vec{b}$

↳ $\vec{a}$ & $\vec{b}$ ARE PARALLEL IF $\vec{a} \times \vec{b} = 0$ ($\theta = 0$ or π)

Triple Products —

$a \cdot (b \times c)$

↳ $\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$ } Scalar Triple Product

↳ geometrically : parallelepiped formed from $\vec{a}, \vec{b}, \vec{c}$

↳ $|\vec{b} \times \vec{c}| = A_{base}$

$V_{parallelepiped} = Ah = |b \times c||a||\cos\theta|$ ↳ if θ b/t $\vec{a}$ and A_{base},

$= |a \cdot (b \times c)|$ $\quad h_{par.} = |\vec{a}||\cos\theta|$

↳ if $= 0$, $\vec{a}, \vec{b}$, and $\vec{c}$ MUST be coplanar

12.5 "Lines & Planes"

Parametric equations —

L is through (x_0, y_0, z_0) and in direction $\langle a, b, c \rangle$

$$\begin{cases} x = x_0 + at \\ y = y_0 + bt \\ z = z_0 + ct \end{cases}$$

↳ any scalar multiple of $\langle a, b, c \rangle$ is the SAME direction as $\langle a, b, c \rangle$ ∴ parallel

Symmetric equations —

↳ eliminate parameter t

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

Planes —

↳ can be fully described by a point lying on it, and the normal vector $\vec{N}$ perpendicular to the plane

vector eqn. of plane: $\vec{N} \cdot (r - r_0) = 0$

or $\vec{N} \cdot r = \vec{N} \cdot r_0$

scalar eqn. of plane:

↳ thru point (x_0, y_0, z_0)

↳ w/ $\vec{N}$: $\langle a, b, c \rangle$

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$

OR

$$ax + by + cz = d$$

↳ some constant

* two planes are parallel IF $\vec{N}_1 \propto \vec{N}_2$

* angle b/t 2 intersecting planes = θ b/t $\vec{N}_1$ and $\vec{N}_2$

→ Intersection of 2 planes:

Find symmetric eqns for intersection of $x + y + z = 1$ and $x - 2y + 3z = 1$.

let $x = \lambda$, $(y + z = 1 - \lambda)(2)$

$+ \; -2y + 3z = 1 - \lambda$

$-5y = -3 + 3\lambda + 1 - \lambda$

$y = \frac{2\lambda - 2}{-5}$

$x = \lambda$

$5z = 2 - 2\lambda + 1 - \lambda$

$z = \frac{3 - 3\lambda}{5}$

$$\left.\begin{aligned} x &= \lambda \\ y &= -\tfrac{2}{5}\lambda + \tfrac{2}{5} \\ z &= -\tfrac{3}{5}\lambda + \tfrac{3}{5} \end{aligned}\right\} \times 5$$

$$L: \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5 \\ -2 \\ -3 \end{pmatrix} \lambda + \begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}$$

$$\boxed{\frac{x}{5} = \frac{2 - y}{2} = \frac{3 - z}{3}}$$

Distance from a point to a plane —

given point $P_1(x_1, y_1, z_1)$ & plane $ax + by + cz + d = 0$,

$$\boxed{D = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}}$$

12.6 "Cylinders & Quadric Surfaces"

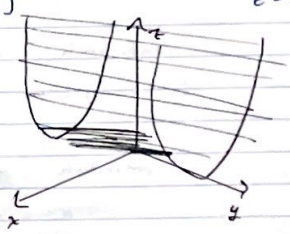
to sketch a graph of a surface, must determine curves of intersection of the surface w/ planes parallel to xy, xz, and yz planes ⟹ TRACES (cross-sections)

→ Cylinders

↳ surface that consists of all lines (rulings) that are parallel to a given line and pass through a plane curve

eg. $z = x^2$

- any plane $y = k \parallel xz$ plane intersects graph in a curve $z = x^2$



* if a variable x, y, or z is missing from the equation of the surface, IT IS A CYLINDER

"parabolic cylinder" ↳

→ Quadric Surfaces
- graph of a second-degree eqn in three variables $x, y,$ and z

general : $Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + Gx + Hy + Iz + J = 0$

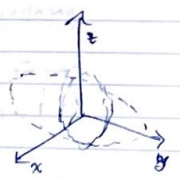
standard : $Ax^2 + By^2 + Cz^2 + J = 0$

OR

$Ax^2 + By^2 + Iz = 0$

Ex. 1

Sketch $x^2 + \frac{y^2}{9} + \frac{z^2}{4} = 1$



ELLIPSOID

substitute $z = k$, $k^2 < 4$

↳ trace in x-y plane : $x^2 + \frac{y^2}{9} = 1$

substitute $x = k$, $k^2 < 1$

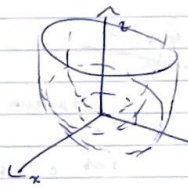
↳ trace in y-z plane : $\frac{y^2}{9} + \frac{z^2}{4} = 1$

substitute $y = k$, $k^2 < 9$

↳ trace in x-z plane : $x^2 + \frac{z^2}{4} = 1$

Ex. 2

Sketch $z = 4x^2 + y^2$



ELLIPTIC PARABOLOID

substitute $z = k$

↳ trace in x-y plane : $4x^2 + y^2 = k$

↳ family of ellipses (↑ radius)

substitute $x = k$

↳ trace in y-z plane : $z = y^2 + 4k^2$

↳ family of parabolas (↑ vertex)

substitute $y = k$

↳ trace in x-z plane : $z = 4x^2 + k^2$

↳ family of parabolas (↑ vertex)

Ex. 3

Sketch $z = y^2 - x^2$

substitute $z = k$

↳ trace in x-y plane : $k = y^2 - x^2$

↳ family of hyperbolas

HYPERBOLIC PARABOLOID

substitute $x = k$

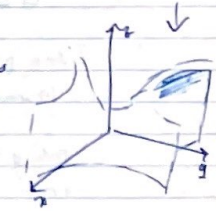
↳ trace y-z : $z = y^2 - k^2$

↳ parabolas opening upwards (↓ vertex)

substitute $y = k$

↳ trace x-z : $z = k^2 - x^2$

↳ parabolas opening downwards (↑ vertex)



Ex. 4

Sketch $\frac{x^2}{4} + y^2 - \frac{z^2}{4} = 1$

sub $z = k$, $k^2 < 4$

↳ trace $x-y$: $\frac{x^2}{4} + y^2 = 1$

↳ ellipses

sub $x = k$, $k^2 < 4$

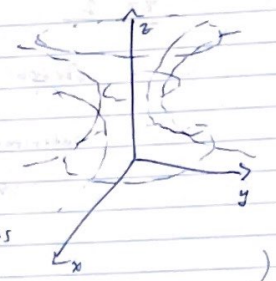
↳ trace $y-z$: $y^2 - \frac{z^2}{4} = 1$

↳ hyperbola

sub $y = k$,

↳ trace $x-z$: $\frac{x^2}{4} - \frac{z^2}{4} = 1$

↳ hyperbola



HYPERBOLOID OF ONE SHEET

Ex. 5

Sketch $4x^2 - y^2 + 2z^2 + 4 = 0$

$$-x^2 + \frac{y^2}{4} - \frac{z^2}{2} = 1$$

sub $z = k$, $k^2 < 2$

↳ trace $x-y$: $-x^2 + y^2 = 1$

↳ hyperbola

sub $x = k$, ~~$k^2 < 4$~~

↳ trace $y-z$: $\frac{y^2}{4} - \frac{z^2}{2} = 1$

↳ hyperbola

sub $y = k$, $k^2 < 4$

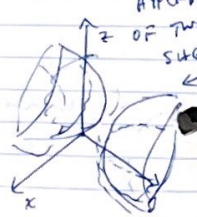
↳ trace $x^2 + \frac{z^2}{2} = -1$

↳ DOESN'T EXIST

$|k| > 2$:

$x^2 + \frac{z^2}{2} = \frac{k^2}{4} - 1$

↳ ellipses



HYPERBOLOID OF TWO SHEETS

4 CONIC SECTION EQUATIONS

CIRCLE	$ax^2 + ay^2 = d$ → constant, ≥ 0 ↳ coefficients are the SAME
PARABOLA	$ax^2 + by = d$ → constant ↳ only ONE variable is squared
ELLIPSE	$ax^2 + by^2 = d$ → constant ↳ diff coefficients, both x and y squared
HYPERBOLA	$-ax^2 + by^2 = d$ → constant ↳ one (+) and one (−) coefficient

GRAPHS OF QUADRIC SURFACES

surface	equation	surface	equation
ELLIPSOID	$x^2 + y^2 + z^2 = 1$ (all traces ellipses)	CONE	$z^2 = x^2 + y^2$ (vertical traces are hyperbolas except k=0)
ELLIPTIC PARABOLOID	$z = x^2 + y^2$ (var. to 2nd power is axis of solid)	HYPERBOLOID (1 SHEET)	$x^2 + y^2 - z^2 = 1$ (axis of symmetry corresponds to negative var)
HYPERBOLIC PARABOLOID	$z = x^2 - y^2$ - horizontal hyperbolas - vertical parabolas	HYPERBOLOID (2 SHEETS)	$-x^2 - y^2 + z^2 = 1$ (two negatives indicate 2 sheets)

13.1

"Vector function": domain $(t) \in \mathbb{R}$
range $r(t) \in V_3$

$$r(t) = \langle f(t), g(t), h(t) \rangle$$

$$= f(t)\,i + g(t)\,j + h(t)\,k$$

↳ LIMITS —

- if $\lim_{t \to a} r(t) = L$, then length and direction of $r(t)$ approach vector L
- a vector function $r(t)$ is continuous at a if —

$$\lim_{t \to a} r(t) = r(a)$$

given $r(t) = \langle f(t), g(t), h(t) \rangle$ ...

$$\lim_{t \to a} r(t) = \left\langle \lim_{t \to a} f(t), \lim_{t \to a} g(t), \lim_{t \to a} h(t) \right\rangle$$

"space curve": set of all points in a vector function

↳ $x = f(t)$
$y = g(t)$
$z = h(t)$

13.2

DERIVATIVES

given $r(t) = \langle f(t), g(t), h(t) \rangle$ ...

$\boxed{r'(t)} \longrightarrow$ "tangent vector"

↳ $\boxed{\langle f'(t), g'(t), h'(t) \rangle}$

$\boxed{\dfrac{r'(t)}{|r'(t)|}} \longrightarrow$ "unit tangent vector"

always points in direction of increasing t

INTEGRALS

given $r(t) = \langle f(t), g(t), h(t) \rangle$ ...

$$\boxed{\int_a^b r(t)\,dt = \left(\int_a^b f(t)\,dt\right) i + \left(\int_a^b g(t)\,dt\right) j + \left(\int_a^b h(t)\,dt\right) k}$$

13.3

length L of a space curve:

$$L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} = \int_a^b |r'(t)|\,dt$$

to parametrize a curve w/ respect to arc length s:

$$\frac{ds}{dt} = |r'(t)| \longrightarrow s(t) \longrightarrow t(s) \longrightarrow r(t(s))$$

r still dependent on t

curvature of a curve:

↳ measure of how quickly a curve C changes direction

↳ magnitude of rate of change of unit tangent vector as s varies

↳ easiest formula for curvature κ —

$$\boxed{\kappa(t) = \frac{|r'(t) \times r''(t)|}{|r'(t)|^3}}$$

$$\boxed{\kappa = \left|\frac{dT}{ds}\right| = \left|\frac{dT/dt}{ds/dt}\right| = \left|\frac{dT/dt}{r'(t)}\right|}$$

$$T(t) = \frac{r'(t)}{|r'(t)|}$$

↳ if plane curve is in terms of x and not parameter t

$$\boxed{\kappa(x) = \frac{|f''(x)|}{[1 + (f'(x))^2]^{3/2}}}$$

unit normal vector —

indicates direction that curve is turning

↳ $\boxed{N(t) = \frac{T'(t)}{|T'(t)|}}$

↳ orthogonal to tangent vector

binormal vector —

↳ perpendicular to tangent and unit normal

↳ unit vector

↳ $\boxed{B(t) = T(t) \times N(t)}$